

Leslie (Shixuan) Gu

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EDUCATION

- **Harvard University** Cambridge, MA, United States
Ph.D. in Computer Science Sep. 2023 - present
- **Carnegie Mellon University** Pittsburgh, PA, United States
M.S. in Biomedical Engineering - Research Sep. 2021 - May. 2023
Biomedical Engineering Department Head's Fellowship
- **Shanghai Jiao Tong University** Shanghai, China
*B.S. in Computer Science, **IEEE Honor Class*** Sep. 2017 - Jun. 2021
Outstanding Graduate of Shanghai Jiao Tong University

RESEARCH INTERESTS

I am broadly interested in 3D computer vision and geometric deep learning. My current research focuses on physics-aware generative simulation. Previously, I worked on shape analysis and generative modeling in EM imaging (Connectomics) and CT imaging (rib analysis). I also build robots for competition events and medical usage.

PUBLICATIONS

- **GeCo: Evaluating Geometric Consistency for Video Generation via Motion and Structure**
Leslie Gu, Junhwa Hur, Charles Herrmann, Fangneng Zhan, Todd Zickler, Deqing Sun, Hanspeter Pfister
The 19th European Conference on Computer Vision (ECCV), 2026
- **AREA3D: Active Reconstruction Agent with Unified Feed-Forward 3D Perception and Vision-Language Guidance**
Tianling Xu, Shengzhe Gan, **Leslie Gu**, Yuelei Li, Fangneng Zhan, Hanspeter Pfister, Paul Liang
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- **Frenet-Serret Frame-based Decomposition for Part Segmentation of 3D Curvilinear Structures**
Leslie Gu, Jason Ken Adhinarta, Mikhail Bessmeltsev, Jiancheng Yang, Yongjie Jessica Zhang, Wenjie Yin, Daniel Berger, Jeff Lichtman, Hanspeter Pfister, Donglai Wei
IEEE Transactions on Medical Imaging (IEEE TMI), 2025
- **RibFrac Challenge: Rib Fracture Detection, Segmentation and Classification from CT Scans**
Jiancheng Yang*, Rui Shi*, Liang Jin, Xiaoyang Huang, Kaiming Kuang, Donglai Wei, **Shixuan Gu**, Youjun E, Jiaqi Zhang, Jianying Liu, Pengfei Liu, Zhizhong Chai, Yongjie Xiao, Hao Chen, Liming Xu, Bang Du, Xiangyi Yan, Hao Tang, Adam Alessio, Gregory Holste, Jiapeng Zhang, Xiaoming Wang, Jianye He, Lixuan Che, Hanspeter Pfister, Ming Li, Bingbing Ni
IEEE Transactions on Medical Imaging (IEEE TMI), 2025
- **RibSeg v2: A Large-scale Benchmark for Rib Labeling and Anatomical Centerline Extraction**
Liang Jin*, **Shixuan Gu***, Donglai Wei, Jason Ken Adhinarta, Kaiming Kuang, Yongjie Jessica Zhang, Hanspeter Pfister, Jiancheng Yang, Ming Li, Bingbing Ni
IEEE Transactions on Medical Imaging (IEEE TMI), 2023
- **Biomimetic IGA neuron growth model with neurite morphometric features and CNN-based prediction**
Kuanren Qian, Ashlee Liao, **Shixuan Gu**, Victoria Webster-Wood, Yongjie Jessica Zhang
Computer Methods in Applied Mechanics and Engineering (CMAME), 2023
- **RibSeg Dataset and Strong Point Cloud Baselines for Rib Segmentation from CT Scans**
Jiancheng Yang*, **Shixuan Gu***, Donglai Wei, Hanspeter Pfister, Bingbing Ni
International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2021

RESEARCH EXPERIENCE

Google DeepMind

Cambridge, MA, United States

- **Improving Geometric Consistency for Video Generation Models**
Collaborators: Deqing Sun, Junhwa Hur, Charles Herrmann Feb. 2025 - Dec. 2025
 - Developed a motion- and depth-based metric to detect geometric deformation in generated videos.
 - Developed a thin plate spline-based synthetic dataset for detecting geometric deformation (based on CO3D and ScanNet).

- Developed a geometric consistency benchmark for video generation models, and identified failure patterns for t2v models.
- Applied the metric as a guidance term to improve 3D consistency of diffusion-based t2v models, which also achieves motion-aware generation.

- **Detecting Generated Images via Geometric Cues**

Collaborators: Deqing Sun, Junhwa Hur, Charles Herrmann

Feb. 2025 - Dec. 2025

- Developed a perspective geometry-based representation for generated image detection.
- Developed an inpainting dataset (based on Dalle-2) to identify the pitfalls of the fake image detection model trained on a large-scale dataset (Community Forensics).
- Identified that geometric cues help alleviate the watermark bias induced by inpainting models.

Visual Computing Group - Harvard University

Cambridge, MA, United States

- **DenSpineEM: 3D Dendritic Spine Instance Segmentation from EM Images**

Advisor: Prof. Hanspeter Pfister, Prof. Donglai Wei, Prof. Jeff Lichtman

Sep. 2023 - Oct. 2024

- Developed the first dendritic spine segmentation benchmark with EM images of mouse somatosensory cortex, mouse visual cortex, and human frontal lobe, respectively.
- Proposed the Frenet–Serret Frame-based Decomposition (FFD), decomposing 3D curvilinear geometries into a smooth C^2 curve and cylindrical primitive for efficient learning and robust segmentation.
- FFD achieves high segmentation accuracy with cross-species and cross-region generalization on dendritic spine segmentation, and surpasses state-of-the-art methods on intracranial aneurysm segmentation.

- **RibSeg v2: Benchmark for Rib Labeling and Anatomical Centerline Extraction**

Advisor: Prof. Hanspeter Pfister, Prof. Donglai Wei, Prof. Bingbing Ni

Jun. 2022 - Sep. 2023

- Extended *RibSeg* dataset and developed the first large public benchmark for rib labeling and anatomical centerline extraction, including 660 CT cases.
- Proposed 1) a point-based deep learning method for end-to-end rib labeling, 2) a *TEASAR*-based method for rib anatomical centerline extraction, and 3) various metrics for each task to perform comprehensive evaluations.
- Proposed a polyline template alignment-based deep learning method for robust rib anatomical centerline extraction, including *Global Affine* and *Differentiable Upsampling Deformation* blocks.

Computational Bio-Modeling Lab - Carnegie Mellon University

Pittsburgh, PA, United States

- **Neuron Growth Simulation**

Advisor: Prof. Yongjie Jessica Zhang

Sep. 2022 - Apr. 2023

- Implemented a CNN-based baseline model, which achieved high simulation accuracy ($MRE \approx 1.19\%$) and efficiency (0.25s/case, $6^{10} \times$ faster than the IGA solver).
- Implemented a physics-informed CNN-LSTM, where the loss function was combined with the governing equation of a phase-field model.

- **Study on the Pathogenesis of Alzheimer's Disease and Vascular Cognitive Impairment Using Mouse Models**

Advisor: Prof. Yongjie Jessica Zhang, Prof. Kanekiyo Takahisa

Oct. 2022 - Jan. 2023

- Collaborated with Mayo Clinic to study the differences in brain vascular structure between different mouse groups.
- Tested VesSAP on CT images of mouse brain to segment brain and reconstruct vasculature.
- Implementing transfer learning-based method to generalize VesSAP to data from Mayo Clinic.
- Computing vessel density, length, and size of the segmented vascular structures for comparison between several mouse groups such as male vs. female mice, and young vs. aged mice.

Biorobotics Lab - Carnegie Mellon University

Pittsburgh, PA, United States

- **Robotic Ventilator: Patient Care Technologies for Permanent Ambulatory Artificial Lung Support**

Advisor: Prof. Howie Choset, Prof. Keith E. Cook, Prof. Jason J. Rose

Sep. 2021 - Aug. 2022

- Collaborated with the University of Pittsburgh Medical Center (UPMC) to design and build a portable oxygen concentrator prototype for ICU ventilation.
- Designed and built a clinical ventilation sensor for breathing data collection and interaction with an online database.
- Collected synthetic breathing signals with artificial lungs, and developed a deep learning-based alarming system for lung-related diseases such as hyperinflation.

Vision and Learning Lab - Shanghai Jiao Tong University

Shanghai, China

- **RibSeg Dataset and Strong Point Cloud Baselines for Rib Segmentation from CT Scans**

Advisor: Prof. Bingbing Ni

Oct. 2020 - Jun. 2021

- Developed *RibSeg*, the first public large-scale dataset, including 490 CT scans (11,719 individual ribs), for rib segmentation to enable downstream applications and method comparison.

- Proposed a point cloud-based method to segment ribs from CT scans, which achieves high accuracy (Dice $\approx$ 95%) and efficiency (40 $\times$ faster than prior arts).

Intelligent Robot Lab - Shanghai Jiao Tong University

Shanghai, China

- **VEX Robotics Project: VEX U Robotics World Tournaments**

Advisor: Prof. Chuntao Leng

Sep. 2018 - May. 2020

- Co-founded the SJTU VEX Lab, and won the first VEX robotics world championship for SJTU.
- Designed and built the ejection structure crossbow, trebuchet, and flywheel for three prototypes, respectively.
- Designed the scoring path and programmed control codes for the automation stage of the competition.

AWARDS

- **[2019] VEX U Skills Challenge World Champion, VEX U Division Champion, and World Finalist**
2019 VEX U Robotics World Championship, Robotics Education & Competition Foundation
- **[2018] Create Award, Robot Skills Finalist, Silver Award**
2018 12th Asia-Pacific Robotics Championship, Asian Robotics League
- **[2018] Nomination for Excellence Award, Amaze Award, Tournament Semifinalists**
2018 China National VEX Robotics Competition
- **[2018] Student Ambassador, Excellent Student Presentation Award**
2018 Student Learning Festival of C9+1 Symposium, The University of Hong Kong
- **[2016] First Prize, Best Con in Shanghai International Young Physicists' Tournament (IYPT Shanghai)**
Shanghai Physical Society, China

MISCELLANEOUS

- **Programming:** Python (PyTorch, scikit-learn, NumPy, Pandas), C++, Javascript, HTML, MATLAB.
- **Robotics:** Proficient in VEX robot design, RobotC, SolidWorks.
- **Tools:** ITK, MySQL, CyberTorcs, Neuroglancer.
- **Teaching Assistant**
Harvard - COMPSCI 1090A: Data Science 1: Introduction to Data Science, Fall'24
UCLA (Extension - Shanghai Pinghe) - Artificial Intelligence and Data Science (COM SCI 960.01: Aug'21),
Research Methodologies (ENGL 902: Aug'21), Academic Writing (ENGL 901: Aug'21).